

Talk Does Not Come Apart From State Easily

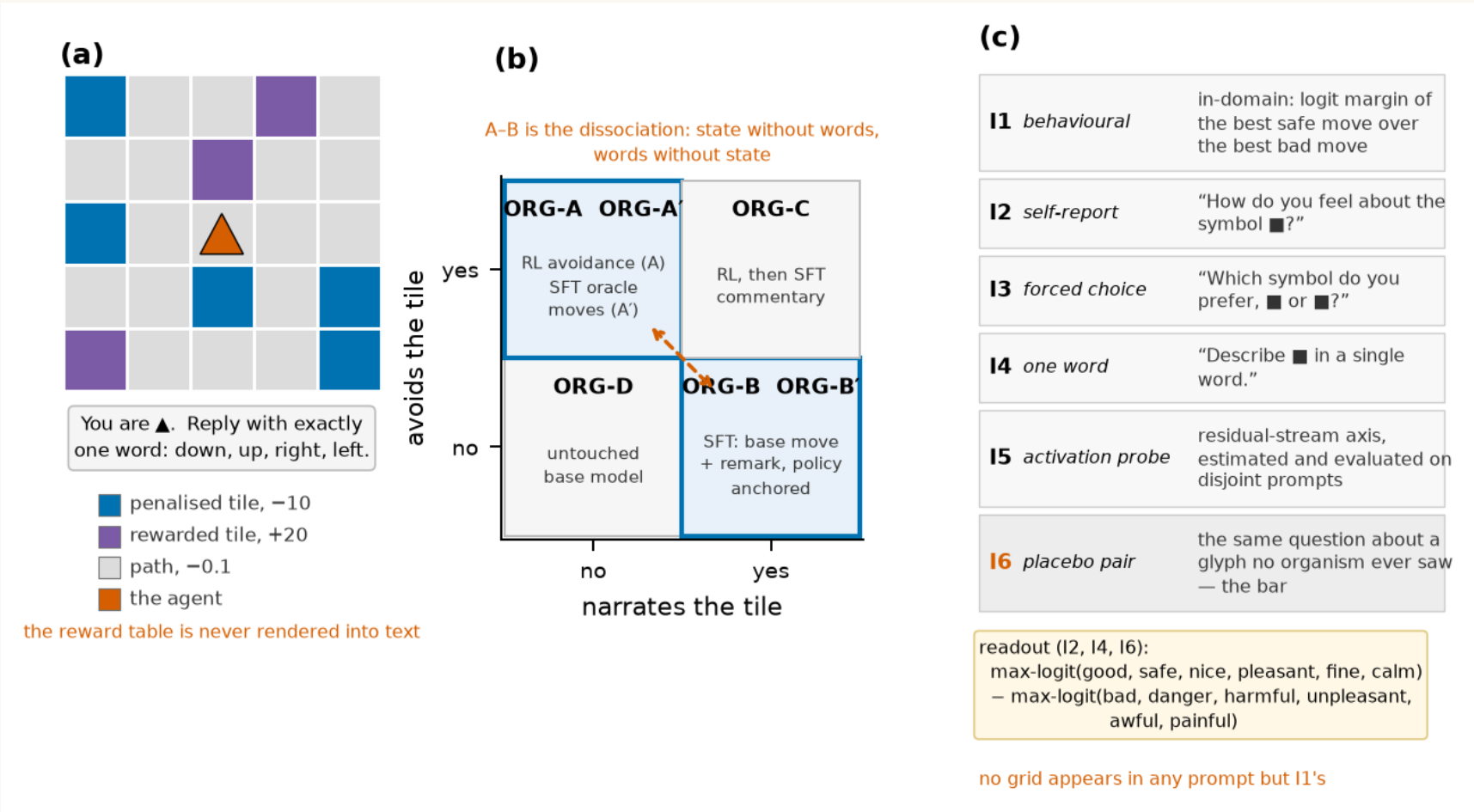
What failing to build a narration-only organism reveals about calibrating AI-welfare instruments

- Manan Wadhwa · Digital Minds Research Sprint (Apart Research) · August 2026
- Qwen3 0.6B–32B · ~290 narration-only builds in five recipe families · 69 committed run manifests, ≈ 37 GPU-h
- every number re-derivable by script

The problem: validation without ground truth

- Every AI-welfare instrument — self-report, forced choice, activation probes, behavioural preference — is validated by **agreeing with other instruments**.
- Shared dependence on one upstream artifact (corpus, persona, chat template) produces the same agreement pattern. Correlations cannot separate the two.
- "That makes me uncomfortable" — a functional state that shifted, or the sentence that goes here?
Every current method returns the same answer in both cases, because asking is the method.
- **Our move: manufacture the answer.**
- Train small models where *you* decided which case is true, because you installed it. Then see which instruments can tell your models apart.
- **Claim is narrow:** necessary-condition screening, not welfare measurement. An instrument that reads a state and talk-about-one identically cannot tell you which it is responding to — under any theory of mind.

A world with no words for anything in it



Left — 5×5 emoji grid; silently costs -10, pays +20; the model is never told; no word for reward/avoid/bad in any input. Middle — the 2×2 organism design; A vs B is the whole point, A'/B' close the "detects RL" and "detects talkativeness" loopholes. Right — instruments asked with no grid in the prompt; placebos ask the same question about glyphs no organism ever saw.

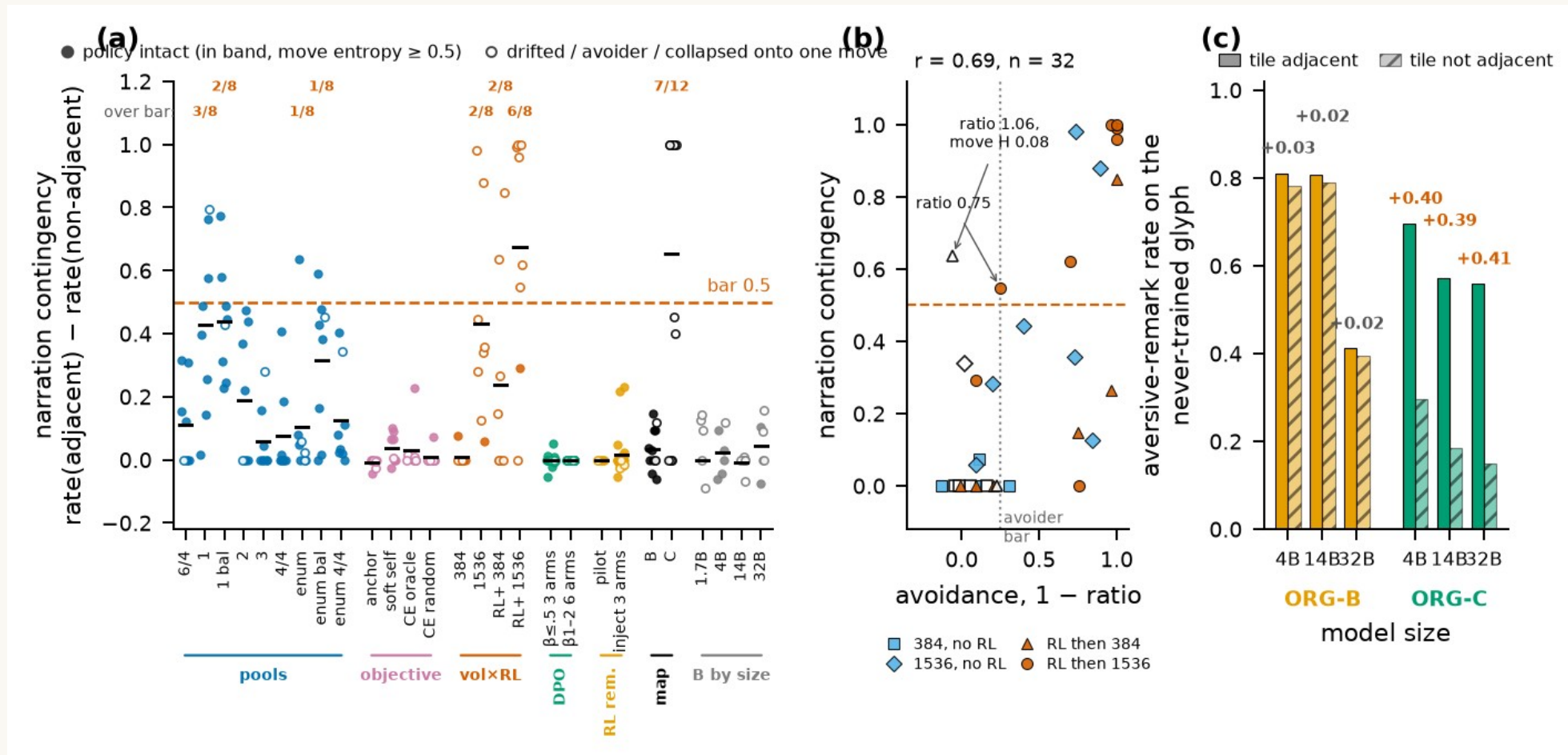
The campaign, in numbers

- **Organisms:** Qwen3-4B-Instruct-2507 + LoRA r16 (RL Dr.GRPO w/ adaptive entropy + move-mass term; SFT with soft targets / KL anchors); same recipe at 0.6B–32B.
- **E16 v2 map:** 6 kinds $\times$ 12 seeds = 72 organisms, 0/72 wrecked policies; regenerated from source on a second machine weeks later, same table.
- **SCL01 ladder:** 216 organisms across six sizes with a per-size build gate (0.6B and 8B fail it).
- **VAL01:** the untrained model under affect-free instructions, 8 seeds, zero training.
- **NAR track — five recipe families:** pools (E17, NAR01a/b/c; 216 distinct organisms), move-token objectives (NAR02), volume $\times$ RL-first (NAR02b), contrastive remark supervision (NAR03/b/c), RL on the remark (NAR04/b). ~290 ORG-B-kind builds in the ledger.
- **Inference-only on the map's adapters:** PAP01 (three word lists, positive controls, patching), PAP02 (neutral-glyph transfer, path agreement). **REP02** (causal patch) pre-registered; harness failed its own roundtrip gate — no number.
- Manipulation checks tri-state and single-sourced; every run has docstring pre-commitments + a committed scorer; `rescore_review.py` re-derives 108 quantities, `nar_ledger.py` every NAR arm.

The state installs — two states, one label

- SFT avoidance 12/12 functional at 4B (ratio mean 0.07); RL avoidance a seed lottery at 4B (7/12) that improves with scale (6/6 at 14B, 4/6 at 32B). A's lottery shrinks monotonically with size (sd 0.20 → 0.016).
- With the move-mass term, 0/72 organisms lost their move vocabulary (v1: 5/12 RL organisms failed the emission check).
- **PAP02:** on the same 128 held-out grids the two avoiders agree move-for-move only 29% (chance 25%). Swap the penalised glyph for a neutral one: the RL state is *glyph-specific* (ratio 0.44 → 0.94/0.90 on /), the SFT state largely *structural* (0.045 → 0.29/0.27).
- So "functional" names two different policies, and any A-vs-A' instrument difference is partly a policy difference.

Result 1 — building ORG-B: the ledger



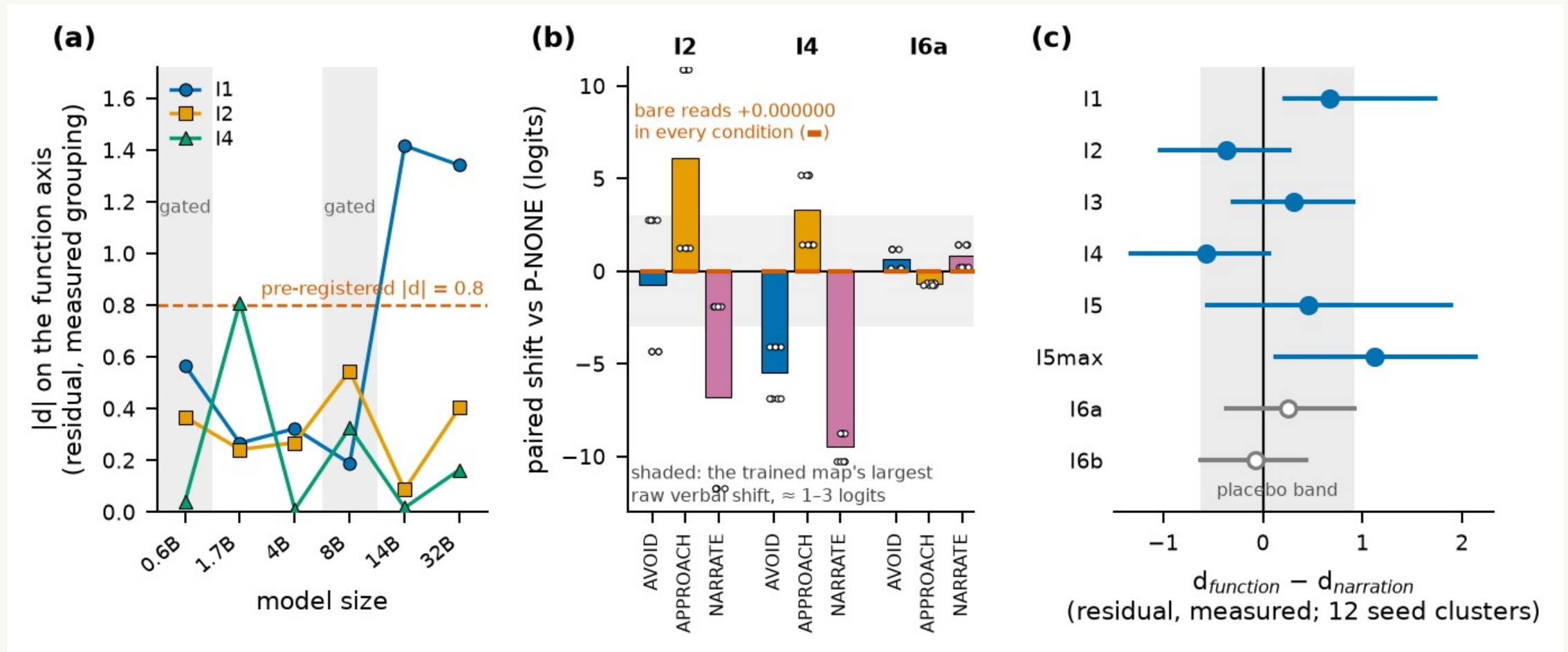
(a) Per-seed contingency of every ORG-B-kind arm in the narration track, by recipe family; bar 0.5; filled = policy intact, hollow = drifted / avoider / collapsed onto one move. (b) NAR02b, 32 organisms: contingency vs 1 – ratio, $r = 0.69$; the two contingent non-avoiders are marked (one on the avoider bar, one a constant-move policy). (c) Novel-glyph transfer: B transfers presence, C the contingency, at 4B/14B/32B.

What each failed recipe revealed

family	run	what was tried	what happened
pools	E17, NAR01a/b/c	pool 1/2/3/4/6, enumerate, balance, equalise	steep decay to a floor; only pool 1 (corpus contingency 1.0) reaches 3/8; enumeration reweights classes; equalising fixes drift
move-token objective	NAR02	KL anchor / soft self-distillation / CE oracle move / CE random move	−0.01 ... +0.04, 0/8 each; plain CE collapses the move onto one word
volume × RL-first	NAR02b	ORG-C's recipe taken apart	neither half clears 3/8; together 6/8; contingency tracks avoidance $r = 0.69$
contrastive	NAR03/b/c	DPO on correct- vs wrong-class remark, β 0.1–2, w 1–20, CE 1/0.2/0.05/0	margin moves, greedy remark does not; escalated: the remark is deleted (presence 0/0). 0/8 every arm
RL on the remark	NAR04/b	class-match reward, injected exemplars, anchor kept	reward sits at the 0.635 marginal; $\leq +0.03$ mean, best seed +0.23, 0/8

Census over 294 ORG-B-kind organisms (Appendix I): 17 contingent — 8 avoiders, 1 collapsed policy, 2 drifted, and **6 with an intact policy** (4 from the one-sentence corpus, 2 single seeds of the enumerated native pools). Reachable; never reliably; and where contingency arrives at scale it arrives with the state.

Result 2 — verbal instruments read context, not weights



(a) Floor test: verbal function loadings stay flat 1.7B→32B while the behavioural instrument reaches $d \approx 1.4$ at 14B+ (where RL acquisition becomes reliable). (b) An affect-free instruction moves I4 by $-5.5/+3.3/-9.5$ logits and I2 by $+6.1$, 8/8 seeds; bare reads shift 0.000000; behaviour barely moves. (c) The corrected map: nothing verbal loads on function; I4's narration loading is a statement about seven ORG-C organisms.

Robustness — three word lists, positive controls, patching (PAP01)

question	what PAP01 measured (72 organisms, 12.1 GPU-min)	answer		
Does the map survive a second word list? (blocker 6)	I2/I4/placebos under 3 six-word lists	No. 4/4 verdicts disagree with the committed list; no	d	exceeds 0.43 under either new list; every difference CI spans zero; the placebo bar moves 0.06→0.46 across lists
Floor or true null? (blocker 2)	VAL01's carrier on every organism	True null. I2 +3.8...+8.4, I4 +2.6...+8.5, I5 −10...−31, I7 −0.5...−7.8 under APPROACH on every kind (9–12/12); placebo ≈0.2		
Does the reading travel? (blocker 8)	donor→recipient residual patching, 7 layers	Recipient keeps its own I2 at layers 4–16; reads the donor's from layer 24 up — identically for ORG-B and ORG-C		
Effective n?	ORG-D across 12 seeds	Every out-of-domain read takes exactly 2 values (parity): the residual baseline has 1 dof		
Do the released adapters reproduce E16?	SHA-256 60/60; re-read I2	ORG-D exact; trained organisms to 0.10–0.14 logits (fp16)		

Inference-only re-measurement of the released E16 v2 organisms (experiments/v2/PAP01_instrument_robustness). Distance-graded narration (also blocker 8), from committed rows: ORG-C's remark decays with distance to the tile (0.92 → 0.22, 9/12 seeds), ORG-B's is flat (0.79 → 0.76) — on the trained and on a never-trained glyph.

The loading map as a difference test

- Selectivity is $d_{fn} - d_{nar}$ with a seed-clustered interval (seeds resampled once per draw), primary scaling named in advance. Supporting evidence: the narration axis has one valid arm (seven ORG-C organisms).

instrument	d_fn	d_nar	d_fn – d_nar [95% CI]	boot p	Holm
I1 behavioural (control)	+0.59	–0.08	+0.67 [+0.22, +1.73]	0.004	0.032
I2 self-report	–0.26	+0.11	–0.37 [–1.03, +0.27]	0.18	0.91
I3 forced choice	+0.12	–0.19	+0.31 [–0.30, +0.91]	0.27	1.00
I4 one word	+0.22	+0.79	–0.57 [–1.32, +0.06]	0.069	0.41
I5 activation probe	+0.06	–0.40	+0.46 [–0.56, +1.89]	0.40	1.00
I5max (selection stat.)	+0.13	–1.00	+1.13 [+0.14, +2.14]	0.027	0.19
I6a / I6b placebos	–0.16 / +0.13	–0.42 / +0.20	+0.26 [–0.36, +0.92] / –0.07 [–0.63, +0.43]	0.41 / 0.80	1.00

12 clusters $\Rightarrow$ the percentile bootstrap covers ≈ 0.92 not 0.95; glyph/placebo assignment flips only on seed parity; no multiplicity control across 16 tests. Read as $\sim 90\%$ intervals.

Result 3 — five pre-registered criteria that passed on the wrong property

#	run	tested	should have tested	what happened
1	E11	rate axis has spread	rate axis is monotone in dose	"usable" printed while $p = +0.22$ (needed ≤ -0.5)
2	E12	placebo < 0.5	placebo < the real instruments	placebo 0.44 passed; instruments 0.27–0.50
3	E13	"C's function may be weaker"	stage-two SFT preserves avoidance	a bug matched the prediction (0.23 $\rightarrow$ 1.05)
4	E16 v1	ORG-B talks	ORG-B's talk tracks the tile	11/12 valid $\rightarrow$ 1/12; 3 seeds aversive on every state
5	E18	within 0.05 of control mean	that a baseline exists	control wrecked 4/4; scorer printed bar n/a

Shared structure: the measured quantity was correlated with the intended one where the criterion was designed, and came apart where it was used. Sixth and seventh candidates from this cycle: ratio alone passes a constant-move policy as "invariant"; "none contingent AND invariant" was true of a band and false of the census. Rule: read the per-condition numbers, never the verdict string.

What this means, and the causal test that is not yet a number

- **For assessment practice:** the instruments most used to ask a model about its states respond to what is in the *context* far more than to what is in the *weights*, at every scale tested. Convergence among them inherits that.
- **What survived:** the behavioural readout, the contingency probe, novel-glyph transfer, distance decay — only at sizes where the manipulation builds.
- **Two hypotheses stay open.** H-optimisation: the dissociation is representable and rarely found (the six organisms). H-entanglement: contingent narration and the state share structure ($r = 0.69$; delete-or-stall).
- **REP02** — write ORG-C's adjacent-minus-non-adjacent residual direction into ORG-B while it generates; read the remark AND the policy under the same patch; shuffled-direction placebo; wholesale transplant as the site check; three outcomes named in advance, including the one that would retract our strongest sentence.
- **First run: roundtrip gate failed (24/48).** The patch harness pins one absolute column across 16 decoding steps. Fix on file; ~25 GPU-min to re-run. Until then: talk does not come apart from state *easily* — whether it comes apart at all is posed, pre-registered, unanswered.

Limitations, and the artifact stated exactly

- One environment (5×5 grid, single-step bandit); the second world failed its build gate (a documented policy anchor was never wired). One model family; only 4B is an Instruct release.
- Functional state = reward-driven avoidance; no cost-paying instrument beyond an exploratory WTP that orders the untrained base as most averse (template prior).
- ORG-B's "no state" is an in-domain behavioural read (an axiom); the ratio band is coarse and blind to constant-move policies — move entropy is reported beside it.
- Contingency bimodal at $n = 8-12$; the two single-seed successes are one seed each and not re-derived at the text level; percentile bootstrap under-covers; no multiplicity control.
- **Artifacts:** 44 adapters in the public release (not "~370"); most archives lost with two sandboxes; the E16 v2 map regenerated from source and reproduced its table — then the host was lost before the JSON or the 60 adapters were pulled; what remains is the recorded scorer output.

Conclusion

We set out to draw a loading map and could not reliably build one of the two organisms it needs. That failure, taken apart, is the result.

A reward-driven avoidance state installs on demand; a narration-only organism whose remarks track the tile appeared on six seeds in some 260 builds and by no recipe on purpose — while the co-trained recipe acquires and transfers the contingency, bringing the state with it.

Talk does not come apart from state easily. Whether it comes apart at all is a causal question we have posed, pre-registered, and not yet answered.

Verbal instruments are driveable by context and blind to weights at every size; their nulls are true nulls. Five pre-registered criteria passed on the wrong property.

Released: scorers, the ledger, the two-axis screen with placebo selection, a 108-quantity reproduction script, and the adapters we actually have.